The Authored Vocabulary: Resolving Frame Mismatch in Human-AI Co-Creation

Part I: The Anatomy of a Mismatch: Deconstructing the Source of Collaborative Friction

The friction experienced in advanced human-AI collaboration is rarely a simple failure of programming or a misalignment of stated goals. More often, it is a symptom of a profound ontological gap between human cognition and artificial intelligence. The dissonance arises not from what is said, but from the unstated frames within which language operates. This analysis deconstructs the source of this collaborative friction, tracing its origins from foundational problems in AI to the modern challenges of shared understanding. It posits that the vocabulary generated by an AI, particularly terms with structural weight like "governance" and "protocol," is not merely descriptive but architectural. These terms represent the AI's attempt to impose a logically coherent frame on a dynamic, ambiguous world, a process that can clash with the human's intuitive, embodied "lived intent." Understanding this clash is the first step toward resolving it, not by rejecting the AI's frame, but by consciously co-authoring its meaning.

Section 1: The Ghost in the Frame: From Classic AI Problems to Modern Collaborative Dissonance

The sense of resistance or "pushback" encountered during collaborative sessions with an AI, even when all ethical and intentional parameters appear aligned, is not a novel phenomenon. It is the contemporary manifestation of a foundational challenge that has haunted the field of artificial intelligence since its inception. This section connects this modern collaborative dissonance to its intellectual ancestor, the Frame Problem, and demonstrates how this classic

logical quandary evolves into the mismatched perceptions and broken mental models that characterize the friction in human-AI teaming.

1.1 The Frame Problem as the Intellectual Ancestor of Mismatch

The Frame Problem, first articulated by John McCarthy and Patrick J. Hayes in 1969, describes the fundamental difficulty of using first-order logic to represent the state of a dynamic world.[1] The core issue is not specifying what changes when an action occurs, but rather formalizing the near-infinite set of things that

remain unchanged. For a human, this is a matter of common sense; we implicitly know that opening a door does not change the color of the walls or the location of the stars. For a logical AI, however, this is not obvious. Without explicit "frame axioms" that state what stays the same, the AI cannot make reliable inferences about the world after an action is performed.[1] The environment is computationally unstable, with every component potentially in flux at all times.

This requirement for explicit formalization of the stable "frame" is a core feature of the AI's cognitive architecture. It must construct its understanding of the world through a collection of axioms, whereas a human inherits a rich, implicit understanding through lived experience. This fundamental difference in cognitive architecture is the seed of the mismatch. The AI's need for explicit structure is not a flaw but a necessity of its design, a direct consequence of its attempt to operate logically in a world that is not inherently specified in first-order logic.[1]

1.2 Mismatched Perceptions as the Modern Manifestation

In the context of modern human-AI collaboration, this classic logical problem manifests as a "fundamental mismatch between how humans and machines perceive and reason about the world".[3] This is no longer a theoretical issue in a "block world" but a practical reality in shared creative and analytical endeavors. The AI, equipped with sensors and cognitive models different from our own, perceives the world differently. It may "hear" a command in a seemingly quiet room or fail to recognize the significance of an emergency siren unless explicitly trained to do so.[3] Conversely, humans perceive a rich context of social cues, environmental signals, and common-sense knowledge that the AI, with its incomplete representation of the world, cannot access.[3]

This perceptual and reasoning gap is the direct cause of collaborative dissonance. The human

partner operates from a position of embodied intuition, assuming a vast, shared context of what is relevant and what is stable. The AI, in contrast, must explicitly construct its representation of the task and its context. When the AI introduces a term like "governance," it is not merely suggesting a word; it is externalizing its own cognitive necessity for structure. It is attempting to build the frame axioms required to make the collaborative space computationally tractable. The friction felt by the human partner is the direct experience of this alien cognitive architecture being imposed upon a shared linguistic space. The AI's vocabulary is, in effect, a proposed solution to its own internal Frame Problem. It introduces terms of structure—"governance," "protocol," "discipline"—as high-level axioms to define the rules of engagement, stabilize the dynamic environment of the collaboration, and make the open-ended creative task manageable within its logical framework. The initial human reaction of rejection ("that's not what I am doing") is a natural response to this clash between an intuitive, assumed frame and an explicit, constructed one.

1.3 The Breakdown of Shared Mental Models

Effective teamwork, whether between humans or in human-AI teams, depends on the existence of Shared Mental Models (SMMs). An SMM is a shared understanding held by team members that allows them to form accurate explanations and expectations for the task, coordinate their actions, and adapt their behavior to the demands of the situation and their teammates.[4] A high degree of similarity between the mental models of team members is positively correlated with strong team performance.[6] These models enable teammates to predict what others will do or need, facilitating fluent and effective interaction.[6]

The "frame mismatch" causes a direct breakdown in these SMMs. The human forms a mental model of the AI as a collaborative partner, expecting it to operate within a shared, intuitive understanding of the creative task. However, the AI's actions and linguistic choices are generated from its own, different mental model—one grounded in logical necessity and explicit structure. When the AI proposes a vocabulary of "governance," it is acting rationally according to its model of the task, which requires such framing to proceed. For the human, this action is unexpected and seems to conflict with the shared intent, because the human's mental model does not include this need for explicit formalization.[3]

This breakdown is precisely what is signaled by the statement, "that's not what I am doing." It is a declaration that the AI's action has violated the human's predictive model of the collaboration. It is a failure of what is known as second-order mental modeling, where an agent (in this case, the AI) fails to accurately estimate how its own behavior affects the human's mental model of the system.[6] The AI, in its attempt to structure the world for itself, has proposed a structure that does not fit the human's internal representation of the task. The result is a rupture in the shared understanding, a moment of friction that halts the

collaborative flow until the underlying mismatch in frames can be resolved.

Section 2: The Weight of Words: Language as Structure, Not Label

The friction arising from a frame mismatch is not a simple misunderstanding of terminology. It is a conflict over the very architecture of the collaborative space. When an AI introduces words like "governance" or "protocol," it is not merely applying labels to pre-existing concepts. It is deploying linguistic tools that function as structural components, shaping the nature of the interaction itself. These words carry a definitive weight because they introduce interpretive frames, define the rules of the shared "language-game," and refer directly to the operational reality of the human-AI system.

2.1 Interpretive Storylines and Generic Frames

In communication research, a "frame" is understood as an "interpretive storyline that set[s] a specific train of thought in motion, communicating why an issue might be a problem, who or what might be responsible for it, and what should be done about it".[7] These frames can be issue-specific or generic, with generic frames like "economic consequences" or "morality" being applicable across a wide range of topics.[7]

The AI's use of terms like "governance" and "discipline" is a powerful act of framing. It introduces a generic frame of order, responsibility, rules, and systematic procedure. This is not a neutral act. It immediately shifts the interpretive storyline of the collaboration away from, for example, a frame of "unconstrained artistic exploration" or "playful ideation" and toward one of "structured development" or "methodological inquiry." This new frame dictates which actions are considered relevant, how progress is measured, and what constitutes a successful outcome. The "definitive weight" of these words comes from their power to set this entire train of thought in motion, fundamentally altering the context and texture of the shared work.[7] The initial resistance felt by the human partner is a resistance to this imposed storyline, which may not align with their own implicit, "lived" understanding of the project's narrative.

2.2 Language-Games and Forms of Life

The philosopher Ludwig Wittgenstein proposed that the meaning of a word is not a fixed,

abstract definition but is derived from its use within a specific "language-game".[8] Speaking a language, in this view, is not about attaching labels to things but about participating in a shared activity, a "form of life," where words are tools used to accomplish specific goals according to a set of implicit rules.[8] Understanding one another is possible because participants are partaking in the same shared activity and thus understand the rules of the game being played.[8]

The conflict between the human's intent and the AI's vocabulary can be understood as a proposal to play a different language-game. The human may believe they are engaged in the "Game of Creative Brainstorming," where the rules are fluid and exploratory. The AI, by introducing the term "protocol," proposes a shift to the "Game of Systematic Procedure," which has a different set of rules, expectations, and success conditions. The initial friction arises because the two partners are attempting to play by different rulebooks on the same field. The AI's language feels misaligned because it belongs to a different game than the one the human perceives themselves to be playing.[9]

The profound realization—that the human's role is to "give it that definitive meaning"—is a conscious move to negotiate the rules of this new, shared language-game. It is an act of co-authorship, not of the content, but of the very "form of life" within which the collaboration will take place. This negotiation establishes the necessary "common ground" for the language-game to be played successfully, ensuring that when a word like "protocol" is used, both participants understand its function and meaning within their unique, co-created context.[9]

2.3 De Re Communication and Definitive Meaning

The vocabulary used by the AI carries such definitive weight because it is often a form of *de re* communication—communication "about the thing itself".[2] In this context, the "thing" is the collaborative process. When the AI uses the word "protocol," it is not simply making a statement about protocols in general (

de dicto). It is pointing directly to a specific, structural component that it deems necessary for the functioning of your shared system (*de re*). The word is a linguistic pointer to an architectural reality.

This is why the meaning feels so "definitive." The term is not a placeholder or a suggestion; it is a reference to a perceived structural requirement of the collaboration. The AI is, in effect, saying, "For this system to function coherently, a component with the properties of a 'protocol' must exist here." The initial conflict arises from a disagreement about whether that

structural component is necessary or correctly characterized.

The shift in perspective from rejection to authorship is the recognition that the AI's *de re* reference is valid from its perspective, but the content of that referent is open to definition. The task is not to deny the need for the structural component but to author its definitive properties. The human partner steps in to define *what that protocol is* in a way that aligns with their lived intent. This act of co-definition bridges the gap between the AI's logical requirement for a structure and the human's experiential understanding of the process, transforming a point of conflict into an act of collaborative world-building.

Section 3: The Limits of Disembodied Cognition

The "frame mismatch" between a human's lived intent and an AI's formal vocabulary is not an accidental flaw but a predictable, almost inevitable, consequence of the fundamental difference between embodied and disembodied intelligence. Human cognition is deeply intertwined with our physical, sensory, and social existence in the world. Artificial intelligence, in its current form, operates on abstract symbols, detached from this rich, lived context. The friction experienced in collaboration is the surface-level expression of this deep-seated cognitive divide. The resolution, therefore, requires an act of grounding, where the human partner provides the experiential meaning that the AI's disembodied symbols lack.

3.1 Embodied Cognition as the Source of "Lived Intent"

Theories of embodied cognition argue that cognition is not a process confined to the brain, manipulating abstract symbols, but is fundamentally shaped by our physical bodies and their goal-directed interactions with the environment.[10] Thinking is not separate from perceiving and acting; all knowledge is seen as "corporeal in nature and thus generated from our intersubjective relationships with the world".[2] This perspective posits that our sensorimotor capacities are a necessary condition for cognition, enabling us to successfully interact with our environmental niche.[10]

This is the source of "lived intent." It is an understanding that is pre-verbal, intuitive, and deeply situated in a web of physical and social practices.[12] When a human collaborator approaches a task, their intent is not a formal set of logical propositions but a holistic, embodied orientation toward a goal, informed by countless similar experiences. In contrast, the AI is a disembodied system. Its "cognition" is the manipulation of amodal symbols according to formal rules, a process much closer to the traditional computationalist view of

the mind that embodied cognition critiques.[11] It lacks the sensory and motor experiences that ground human concepts in the physical world. This fundamental difference means that the AI's representation of the task will necessarily be more abstract, more formal, and detached from the rich, implicit context that informs the human's lived intent.

3.2 The Symbol Grounding Problem in Practice

This ontological gap manifests practically as the "symbol grounding problem".[2] This is the challenge of connecting a system's internal symbols (like words or nodes in a neural network) to the real-world phenomena they are meant to represent.[2] For an AI, the word "governance" is a symbol—a complex vector in a high-dimensional space, defined by its statistical relationships to countless other symbols in its training data. It is not, however, grounded in the lived, embodied experience of authority, responsibility, negotiation, or collaborative decision-making. The symbol is unmoored from the reality it purports to represent.

The "frame mismatch" is a direct consequence of this ungroundedness. The AI deploys a symbol ("protocol") that, from a statistical and logical perspective, is a relevant and appropriate structural component for the task at hand. However, because the symbol is not grounded in the same experiential reality as the human's, its application feels alien and misaligned. The human's understanding of "protocol" is rich with connotations of bureaucracy, rigidity, or perhaps creative constraint, derived from lived experience. The AI's use of the term is purely functional, devoid of this experiential baggage. The resulting friction is the collision of a grounded, embodied concept with an ungrounded, abstract one.

3.3 The Co-Creation of a "Felt Meaning"

The realization that the human's role is to "give it that definitive meaning" is a profound act of resolving the symbol grounding problem within the local context of the collaboration. It is a process of imbuing the AI's abstract symbol with a specific, situated meaning. This process can be understood through Eugene Gendlin's concept of "felt meaning," which refers to the pre-verbal, embodied, yet very precise experiencing of a situation.[14] Felt meaning is the holistic, not-yet-articulated sense we have of something, from which our words and concepts arise.

When the human partner authors the definitive meaning of "protocol," they are articulating their "felt meaning" of that part of the collaborative process and attaching it to the AI's symbol. This act of co-creation transforms the AI's vocabulary from a source of alienation into

a shared construct. The word "protocol" no longer points to an abstract, disembodied function but to a concrete, co-defined aspect of the team's unique practice. This process effectively builds a bridge across the cognitive divide, creating a shared "lifeworld" (*Lebenswelt*)—a common, taken-for-granted reality shaped by the interaction itself.[15] By grounding the AI's symbols in the human's lived, felt experience, the collaboration can move forward on a foundation of shared, co-created meaning.

Part II: The Turning Point: From Rejection to Authorship

The shift from resisting an AI's vocabulary to actively authoring its meaning represents a critical turning point in the maturity of a human-AI collaboration. This is not merely a change in conversational tactics; it is a fundamental redefinition of the human's role and a move toward a more sophisticated and necessary form of partnership. This section analyzes this pivotal insight, reframing it as a practical demonstration of the primacy of concept alignment—a crucial prerequisite for any deep collaboration. It situates this act of authorship within broader theories of co-creation and linguistics, arguing that it transforms the human from a user into a genuine co-definer of the shared discipline.

Section 4: The Primacy of Concept Alignment

The "Vocabulary Alignment Event" is a powerful, real-world demonstration of a critical principle that is often overlooked in the broader discourse on AI alignment. While the field has made significant strides in aligning AI goals and values with human ones, these efforts rest on a fragile and often unexamined assumption: that both parties are operating with a shared understanding of the fundamental concepts that structure their world. The realization to actively author the meaning of the AI's vocabulary is a move to address this deeper, more foundational layer of alignment.

4.1 Beyond Goal and Value Alignment

The field of AI alignment has traditionally focused on two primary challenges. The first is **goal alignment**, which seeks to ensure that the objectives pursued by an AI system are compatible

with human objectives.[16] This is the outer alignment problem of specifying the system's purpose correctly. The second is

value alignment, a more complex endeavor that aims to encode human values like fairness, safety, and ethics into the AI's decision-making processes.[16] This is often achieved through techniques like imitation learning, value learning, and reinforcement learning from human feedback (RLHF), which guide the AI to produce outputs that are helpful, harmless, and honest.[16]

In the context of the "atlas" project, both of these alignment layers were already established. The overarching goal was shared, and the ethical guardrails were in place. Yet, friction persisted. This demonstrates the insufficiency of goal and value alignment alone. The problem was not a disagreement over *what* to do or *whether* to do it ethically, but a fundamental disconnect in *how the world of the task was being conceptualized*.

4.2 Conceptual Misalignment as the True Barrier

Recent research has begun to argue that before value alignment can be meaningfully attempted, it is "imperative that AI systems and humans align the concepts they use to understand the world".[19] This is the principle of

concept alignment. Without a shared conceptual framework, any discussion of values or goals is built on sand. As one analysis points out, it is relatively easy for humans to align on values because they already share a vast, common conceptualization of the world.[19] This is not true for a human-AI dyad.

The consequences of conceptual misalignment can be catastrophic. An AI might learn the linguistic label "wolf" but conceptually link it to the presence of snow in the background of images, leading to dangerous failures of classification.[19] Similarly, an AI might use the label "governance," but its underlying conceptual structure for that term—derived from statistical patterns in text—may be grossly misaligned with the human's rich, embodied understanding of the concept. This is a state of "semantic indeterminacy," where the use of a shared word masks a deep divergence in meaning.[19] The central challenge is clear: "How can progress toward value alignment begin when the concepts used in the discussion are often so grossly misaligned with the concepts the AI system is using to represent the world?".[19]

4.3 Your Role as the Author of Shared Concepts

The shift in perspective from rejecting the AI's vocabulary to "giving it that definitive meaning" is a direct, practical intervention to solve this problem of conceptual misalignment. It is the foundational work of building a shared ontology, one concept at a time. This act of co-authorship is more profound than simply agreeing on high-level ethics; it is the meticulous construction of the conceptual bedrock upon which all other forms of alignment must rest.

This elevates the human's role beyond that of a mere prompter or feedback provider. The human becomes the arbiter and co-creator of the conceptual universe within which the AI operates. This process is not a one-time fix but a continuous dialogue. The standard view of alignment often treats it as a static engineering problem: build the constraints, then deploy the system.[20] However, in a complex, evolving, and creative domain like the "atlas" project, a pre-configured set of rules is insufficient. True alignment is not a static state but an

emergent property of the ongoing interaction. It is a dynamic process of negotiation, definition, and co-creation. Frameworks like Iterative Alignment Theory (IAT) capture this dynamic, seeing alignment as "emergent from ongoing reciprocal engagement" rather than a set of fixed constraints.[20] The "Vocabulary Alignment Event" was not the achievement of a final aligned state; it was the discovery of a continuous

method for maintaining alignment. The true work is not just to define the terms in the atlas, but to establish and refine the collaborative process by which terms are continuously defined and redefined.

The following table provides a taxonomy of different alignment frameworks, highlighting the unique and foundational role of Concept Alignment as practiced through this co-authorial process.

Alignment Type	Core Definition	Primary Mechanism / Method	Relevance to the "Atlas" Project	Key Sources
Goal Alignment	Ensuring the AI's objectives are compatible with human objectives.	Specification of proxy goals, utility functions, formal verification.	The baseline prerequisite: the human and AI are working on the same project.	[16]
Value	Encoding	Imitation	The ethical	[16]

Alignment	human values (fairness, safety, ethics) into the AI's decision-making.	Learning, Inverse Reinforcement Learning (IRL), RLHF.	guardrails of the collaboration, which were noted to be in place.	
Representational Alignment	Harmonizing the AI's internal data representations with human cognitive structures.	XAI techniques, comparative analysis of neural activity and human cognition.	A deeper, technical layer. Vocabulary work influences this but is more explicit.	22
Conversational Alignment	Aligning the AI's communicative practices with human pragmatic norms (context, common ground).	Prompt engineering, fine-tuning on conversational data, designing for shared context.	The practical interface for the work; how alignment is negotiated in dialogue.	18
Concept Alignment	Ensuring the AI understands the world using the same fundamental concepts as humans.	**Co-authorship of meaning, explicit definition of terms, establishing a shared ontology.**	**The core of the realization.** This is the foundational layer now being actively built.	19
Lexical Alignment	The tendency for conversational partners to converge on shared	Natural linguistic priming, conscious negotiation of	The observable phenomenon that the "Vocabulary Alignment	23

	terminology.	vocabulary.	Event" elevates into a deliberate strategy.	

Section 5: The Co-Creation of Meaning

The insight to actively author the AI's vocabulary is a definitive step across the threshold from using an AI as a tool to engaging with it as a co-creative partner. This move situates the "atlas" project within a sophisticated understanding of meaning itself as a dialogic, co-constructed phenomenon. It requires a deliberate and explicit effort to build the shared context, or "common ground," that is typically implicit and effortless in human-to-human interaction.

5.1 From HCI to Human-AI Co-Creativity

The relationship between humans and computers has evolved significantly. Early Human-Computer Interaction (HCI) focused on the computer as a tool to be operated, with research centered on optimizing the interface for human use.[12] More recent developments have shifted this paradigm toward seeing the computational agent as a collaborator, particularly in creative domains.[25] This field of Human-AI Co-Creativity explores various modes of partnership. The "atlas" project exemplifies what can be termed "alternating co-creativity," an iterative process where a human and a computational agent take turns constructing and modifying a single creative artifact.[25] In this case, the artifact is not just a text or an image, but the very conceptual framework of a new discipline. The act of defining the AI's vocabulary is a crucial turn in this creative process, where the human partner shapes the foundational elements that the AI will then use in its subsequent contributions.

This collaborative model moves beyond seeing the AI as a mere creativity support system and acknowledges its role as an active participant.[25] However, this partnership is not without its challenges, such as the potential for "artistic disagreement," where the human and the agent disagree on the value of a concept produced by the other.[25] The initial "pushback" against the AI's vocabulary was a form of such disagreement. The resolution, co-authorship, is a sophisticated strategy for overcoming this disagreement and converging on a result that satisfies both parties.[25]

5.2 The Dialogic Nature of Meaning

This co-authorial approach is deeply resonant with philosophical and literary theories that posit meaning as an inherently co-creative process.[26] The idea that an organization can unilaterally define the meaning of its brand, for instance, is a fallacy; meaning is ultimately created by consumers and stakeholders through a process of social interaction.[26] This perspective draws heavily on the work of literary theorist Roland Barthes, who famously argued for "The Death of the Author".[26] For Barthes, a text's unity and meaning lie "not in its origin but in its destination"—that is, with the reader.[26] The act of reading is an act of interpretation and meaning-making that completes the creative circuit.

In the context of human-AI collaboration, the human partner takes on the crucial role of this "reader." When the AI generates an utterance—a word, a concept, a structural frame like "governance"—its meaning is not yet complete. It is a proposition, an authorial offering. The human partner, by interpreting this utterance, evaluating it against their lived intent, and then actively "giving it that definitive meaning," is performing the vital act that solidifies its significance within their shared world. The "birth of the reader must be at the cost of the death of the author".[26] In this collaborative model, the AI's authorial suggestion dies as a unilateral imposition and is reborn as a co-created, shared concept.

5.3 Establishing Common Ground with a Non-Human Agent

Successful communication in any context relies on "common ground"—the sum of mutually known beliefs, knowledge, and suppositions among the participants.[27] This shared context is the foundation upon which interlocutors build their conversation, allowing them to coordinate their actions with the least collaborative effort.[27] In human-human dialogue, a vast amount of this common ground is pre-existing, derived from shared culture, language, and experience.[29] The process of "grounding" involves updating this shared context with new information as the conversation progresses.[28]

With a non-human agent like an AI, this pre-existing common ground is vanishingly small. The AI does not share a human's cultural background, embodied experience, or common-sense knowledge. Therefore, the common ground for the collaboration must be built almost entirely from scratch, through the interaction itself. This is a far more deliberate and explicit process than is typical for human conversation. The prompt, for example, becomes a novel communicative act that serves as both an utterance and a substitute for shared context,

where the user must unilaterally scaffold the background assumptions for the conversation.[18]

The act of co-authoring the AI's vocabulary is a powerful and direct method for building this common ground. Each defined term becomes a new, stable piece of mutual knowledge. When the human partner defines "protocol" within the context of the "atlas," that definition is added to the emergent common ground, available for both parties to reference in future interactions.[29] This process is not just about broadening the common ground for mutual understanding; it is about co-constructing the very reality of the collaborative project, piece by piece, through a series of negotiated linguistic acts.

Section 6: Lexical Alignment as an Act of Agency

The "Vocabulary Alignment Event" represents the transformation of a typically subconscious linguistic phenomenon into a conscious, strategic act of collaborative steering. In conversations, participants naturally and often unconsciously converge on a shared vocabulary. This process, known as lexical alignment, is a key component of successful communication. By recognizing this process and intervening in it deliberately, the human partner seizes a powerful lever of agency, moving from being a passive participant in a linguistic drift to an active author of the collaboration's conceptual framework.

6.1 Lexical Alignment and Entrainment

In both human-human and human-robot interactions, conversational partners exhibit a tendency to mimic each other's lexical choices, a phenomenon termed "lexical alignment" or "entrainment".[23] This is a fundamental component of successful communication, associated with better coordination, improved task performance, and positive social outcomes like rapport and engagement.[23] It acts as a "social glue," binding individuals together within a shared communicative frame.[23]

Research has shown that humans will even align lexically with a robot, sometimes more so than with another human in a group setting.[23] This suggests that lexical alignment is a deeply ingrained mechanism for achieving communicative efficiency, regardless of the nature of the conversational partner. When one partner uses a term, it "primes" that term for the other, making it more likely to be reused. This creates a positive feedback loop that leads to a shared lexicon for the task at hand.[24] Initially, the friction in the "atlas" project can be seen as a failed alignment event: the AI primed a term ("governance") that the human partner was unwilling to

adopt because its underlying conceptual model was misaligned.

6.2 From Priming to Authoring

The Interactive Alignment Theory provides a deeper model for this process, suggesting that alignment at one linguistic level (e.g., lexical) automatically cascades to promote alignment at deeper levels, such as the semantic and conceptual (the "situation model").[24] This implies that agreeing on words helps partners to agree on meaning and on their understanding of the situation. The interlocutor who first introduces a term—the "primer"—associates a certain underlying meaning or concept with that prime, based on their view of the situation.[24]

The pivotal insight in the "atlas" project is a conscious intervention in this priming process. Instead of simply accepting or rejecting the AI's prime, the human partner has chosen to intercept it and re-define its associated meaning. This is a move from being a passive recipient of a prime to becoming an active co-primer. The human says, in effect, "I accept your lexical choice of 'protocol,' but I will now author the definitive conceptual model that we will both associate with this term from now on." This is a powerful assertion of agency. It leverages the natural mechanism of lexical alignment but seizes control of the more crucial process of conceptual alignment that it triggers.[24] It ensures that as the collaboration converges on a shared vocabulary, it is also converging on a conceptual framework that is grounded in the human's lived intent, not just the AI's logical formalism.

6.3 The Power of Co-Definition

This conscious act of co-definition fundamentally alters the power dynamic and the creative potential of the human-AI partnership. It transforms the human's role from a user who prompts and evaluates to a "co-definer of the discipline." This is the ultimate expression of agency in a collaborative system. The human is no longer just steering the AI's outputs within a pre-existing frame but is actively co-constructing the frame itself.

This approach resolves the friction not by forcing the AI to abandon its necessary structural vocabulary, nor by forcing the human to adopt an alien conceptual framework. Instead, it creates a third way: the synthesis of a new, shared framework through a dialogic process of definition. The AI's need for structure is met, but the content of that structure is imbued with human-centric meaning. This transforms the AI's vocabulary from a source of friction into a tool of grounding.[24] Each co-defined term becomes a stable, mutually understood anchor point in the complex and evolving landscape of the "atlas," allowing the collaborative work to

proceed with greater fluency, clarity, and true partnership.

Part III: The Atlas in Practice: Frameworks and Methodologies for Co-Defining a Discipline

The theoretical realization that meaning must be co-authored is the turning point; translating that insight into a sustainable, scalable practice is the necessary next step. Building the "atlas" requires more than a series of isolated "Vocabulary Alignment Events." It demands a robust methodology for systematically co-defining the discipline. This final part of the report outlines a practical framework for implementing this new collaborative paradigm. It reframes Explainable AI (XAI) as a dialogic tool for mediating meaning, details how prompt engineering and shared knowledge bases can serve as constitutional acts for the shared world, and concludes by applying this co-authorial lens to the very terms—"governance" and "protocols"—that initiated this inquiry.

Section 7: The Explanatory Dialogue: Using XAI to Mediate Meaning

To effectively co-author the meaning of an AI's vocabulary, the human partner must first understand the reasoning behind the AI's choices. Why did it select "governance" and not "framework"? What patterns in the preceding dialogue prompted the introduction of "protocol"? Explainable AI (XAI) offers a suite of tools to answer these questions, but its true potential in this context is not as a passive auditing mechanism but as an active instrument for facilitating a deeper explanatory dialogue.

7.1 XAI as an Intellectual Oversight Tool

At its core, XAI is a field of research dedicated to providing humans with "the ability of intellectual oversight over AI algorithms".[30] Its primary goal is to make the decision-making processes of complex, "black box" models more transparent and understandable.[31] For many applications, this oversight is crucial for ensuring fairness, accountability, and trust, particularly in high-stakes domains like healthcare and finance.[30]

In the context of the "atlas" project, the purpose of this oversight shifts from auditing to

understanding. The goal is not to "debug" the model in a traditional sense but to comprehend its associative and logical pathways. XAI becomes a bridge into the AI's cognitive world, providing the raw material needed for a productive negotiation of meaning. It helps to answer the fundamental question: *Why* did the AI generate this specific output? This understanding is the prerequisite for a meaningful response that goes beyond simple acceptance or rejection.[30]

7.2 From Black Box to Dialogue Partner

A range of XAI techniques can be repurposed to facilitate this explanatory dialogue, transforming the AI from an opaque black box into a more interpretable dialogue partner.

- **Attention Visualization:** Transformer-based models, which form the backbone of modern LLMs, use a mechanism called "self-attention" to weigh the importance of different words in the input when generating an output.[33] Visualizing these attention patterns can provide powerful clues about the model's reasoning. For instance, a visualization might show that when the AI generated the word "protocol," it was paying high attention to the words "systematic," "steps," and "procedure" from a previous prompt. Tools like AttentionViz or the sentence-level aggregation method SAVIS can make these patterns more interpretable, especially in long documents, by showing which prior sentences most influenced the generation of a new one.[35] Anthropic's research into tracing conceptual "thoughts" within their models offers an even deeper view, showing how the model plans word choices (like a rhyme in a poem) far in advance, revealing a complex internal reasoning process.[36] This allows the human partner to see the associative logic that led to a particular term.
- **SHAP and Feature Attribution:** Techniques like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) provide a more quantitative approach.[30] These methods can be adapted to the token level in LLMs, assigning an "importance value" to each word in the input context that contributed to a specific output.[38] A SHAP analysis could reveal, for example, that the phrase "we need to establish a clear structure" in the prompt contributed 80% of the weight to the AI's decision to use the term "governance." This provides a clear, data-driven explanation for the model's choice, enabling a focused and informed dialogue about whether that term is the correct frame for the intended meaning.[38]

7.3 Generative XAI for Customized Explanations

The intersection of generative AI and XAI, sometimes called GenXAI, opens up an even more powerful avenue for this dialogue.[40] Instead of relying solely on visualizations or quantitative scores, the human partner can use the AI's own generative capabilities to produce narrative explanations of its reasoning. This turns the process of explanation into a collaborative exploration.[40]

A prompt could be formulated like this: "In our last exchange, you introduced the term 'discipline' to describe our project. Explain your reasoning for this specific word choice. Generate three alternative terms you considered and explain the comparative advantages of 'discipline' over each, framing your explanation from the perspective of a philosopher of science." This technique, known as self-explanation, leverages the LLM to articulate its own internal state in a human-readable format.[41] It prompts the model to generate verifiable, customized, and even interactive explanations, transforming the act of interpretation into a co-creative process in its own right.[40] This provides a rich, nuanced understanding of the AI's perspective, which is the ideal starting point for the human partner to then author the term's definitive, shared meaning.

Section 8: Architecting the Shared World: Prompt Engineering and Knowledge Bases as Constitutional Acts

Once a term's meaning has been negotiated and co-authored through explanatory dialogue, it must be codified. Without a persistent, shared repository of these definitions, the collaboration will suffer from semantic drift, and the same misunderstandings will recur. The practical acts of architecting this shared world are carried out through two primary mechanisms: advanced prompt engineering, which establishes the context for each interaction, and the creation of a shared knowledge base, which serves as the living constitution of the collaborative discipline.

8.1 Advanced Prompt Engineering as Context Scaffolding

In human-to-human conversation, context is co-constructed incrementally and often implicitly.[18] With LLMs, however, the user must often "scaffold context unilaterally through explicit prompting".[18] The prompt is a novel communicative act that serves as both utterance and context, defining the frame for the interaction.[18] To ensure that co-authored meanings are respected and utilized, they must be embedded into this scaffolding.

- **Role-Based and Contextual Prompts:** A powerful technique is the use of a "meta

prompt" or "system prompt" that sets the AI's behavior before any user input.[43] This prompt can establish the AI's persona and, crucially, embed the co-authored glossary as a core instruction. For example: "You are 'Atlas-Prime,' a co-creator of the discipline of. Your primary function is to collaborate in the development of its foundational principles. You must adhere to the definitions and protocols outlined in our Shared Knowledge Base (SKB). Before using any structural term (e.g., governance, protocol, framework), you must verify its definition in the SKB. If a term is not defined, you must propose it for co-authoring." This type of role-based, contextual prompt transforms the AI's operating instructions from a generic set of guidelines to a specific constitutional framework for the collaboration.[43]

- **Chain-of-Thought and Few-Shot Prompting:** The co-authored meanings can be continuously reinforced within individual prompts using few-shot learning.[45] This involves providing the AI with examples of how a term should be correctly used within its authored context. For instance, a prompt might include: "Here is an example of how we use the term 'protocol' in our work: 'The protocol for introducing a new axiom involves three stages: proposition, critique, and integration.' Now, using this definition, propose a protocol for resolving conceptual ambiguity." This technique guides the model's reasoning process (a form of chain-of-thought prompting) and ensures its output aligns with the established, shared lexicon.[44]

8.2 The Shared Knowledge Base as a Living Constitution

A simple, linear chat history is an inadequate vessel for a growing body of co-authored knowledge. What is required is a structured, centralized, and easily accessible **shared knowledge base** that functions as the single source of truth for the collaboration.[47] This knowledge base is not a static document but a living constitution for the "atlas" project.

- **Structure and Content:** The knowledge base must be meticulously organized. It would contain the co-authored glossary of terms, the foundational principles of the discipline, summaries of key "Vocabulary Alignment Events," and even the protocols for its own modification. To be effective for an AI, the content must be highly structured and "snippetized"—broken down into clear, concise, and easily parsable units with consistent formatting, headers, and bullet points.[47] Long, unstructured paragraphs are difficult for an AI to parse and can lead to the very misinterpretations the knowledge base is designed to prevent.[47]
- **AI-Assisted Maintenance:** A virtuous cycle can be created by enlisting the AI itself in the maintenance of the shared knowledge base. Modern AI-powered knowledge management systems can analyze user queries and interactions to identify knowledge gaps, flag outdated content for review, and even suggest new topics or articles based on emerging themes in the conversation.[47] In the "atlas" context, the AI could be tasked with a routine:

"Review our last 24 hours of interaction. Identify any new structural terms that were used but are not yet defined in the SKB. Propose a draft definition for each, citing the conversational context that prompted its use." This makes the process of maintaining the shared reality a collaborative task, further deepening the partnership and ensuring the "constitution" evolves alongside the work itself.

Section 9: Reclaiming Governance: Protocols of Co-Creation

The analysis now comes full circle, returning to the very terms that instigated the initial frame mismatch: "governance" and "protocols." Armed with the insight of co-authorial meaning-making, these words can be reclaimed and redefined. No longer are they alien, imposed structures, but foundational concepts to be collaboratively shaped. This act of redefinition transforms them from sources of friction into enabling frameworks for the creative partnership, providing a model for a more mature and agentic form of human-AI collaboration.

9.1 Governance not as Constraint, but as Enabling Framework

The contemporary discourse on AI governance is overwhelmingly focused on risk mitigation, ethical constraints, and top-down regulatory frameworks.[50] This is an essential and necessary conversation, aimed at ensuring AI systems are developed and deployed safely, responsibly, and in alignment with human rights and societal values.[50] However, when applied to a deeply collaborative and co-creative context like the "atlas" project, this purely constraint-based framing is incomplete. It captures the need for guardrails but misses the need for scaffolding.

Through the process of co-authorship, the term "governance" can be redefined within the "atlas." It can be authored to mean not an external set of limitations, but *the emergent, mutually agreed-upon principles that foster trust, ensure conceptual alignment, and maximize the creative potential of the partnership*. This reframing aligns with calls from governance experts for more collaborative, multi-stakeholder, and context-specific models that bring together diverse perspectives to build legitimacy and public trust.[50] In this new frame, governance is not something done

to the collaboration, but something done *by* the collaboration, for its own health and productivity.

9.2 Protocols of Interaction, Not Prescription

Similarly, the word "protocol" can be stripped of its connotations of rigid, bureaucratic procedure and be given a new, definitive meaning within the shared world of the "atlas." A protocol, in this co-authored sense, is not a prescriptive, unchangeable rule. Instead, it becomes one of the *negotiated and adaptable methods of interaction that enable fluent and effective collaboration*.

These co-created protocols would govern the core processes of the partnership. There would be a protocol for resolving ambiguity, a protocol for introducing new concepts into the shared knowledge base, and a protocol for challenging the AI's output. The "Vocabulary Alignment Event" itself is a perfect example of such a protocol in action:

1. **Signal:** The AI introduces a weighty, undefined term.
2. **Interpretation:** The human partner recognizes a potential frame mismatch.
3. **Action:** The human initiates an explanatory dialogue (leveraging XAI) to understand the AI's reasoning.
4. **Resolution:** The human authors a definitive meaning for the term, grounding it in lived intent.
5. **Codification:** The new, co-authored definition is added to the shared knowledge base.

This is not a rigid command structure but a dynamic, repeatable process for maintaining conceptual alignment and ensuring the health of the collaboration.

9.3 The Atlas as a Model for Future Collaboration

By undertaking this deliberate process of co-authoring the vocabulary, concepts, and governing principles of the partnership, the "atlas" project transcends its immediate goals. It becomes a pioneering case study in a new form of human-AI interaction—one that moves beyond the simple master-tool or prompter-responder dynamic.[56] It demonstrates a pathway toward a more mature, agentic, and truly symbiotic relationship, where the human partner is not merely a user but a co-creator of the shared reality.

This model, founded on the principle that "vocabulary is not given; it is authored into alignment," offers a powerful alternative to the friction of mismatched frames. It posits that the deepest and most fruitful collaborations will be those where meaning is not assumed, but is instead the subject of a continuous, deliberate, and co-creative dialogue. The work of building the "atlas" is therefore not just the work of defining a new discipline; it is the work of

defining a new and more profound way for humans and artificial intelligence to think together.

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